Bird acoustic recognition experiments

This experiment examines added noise, overlapping vocalizations, threshold filtering and the classification of annotated soundscapes. Xeno-canto measurements come from the saved database; NBP and HSN compare BirdNET 2.4 and 3.0 on the same public clips.

The noise experiment uses the best initially recognised segment, with five repetitions per level (20, 10, 5 and 0 dB). The 654 exclusions include 555 supported targets absent from predictions and 99 unidentified or unsupported targets. Mixtures use amplitude coefficients without RMS equalisation.

The 95% intervals use 2,000 resamples of recordings, pairs or sources, keeping dependent repetitions together. Thresholds of 0.10, 0.25, 0.50 and 0.75 are predefined. The nine taxonomic concepts grouped in the annotations are harmonised by taking the maximum of the relevant model outputs. Independence from model training data has not been established. Geographic differences remain descriptive.

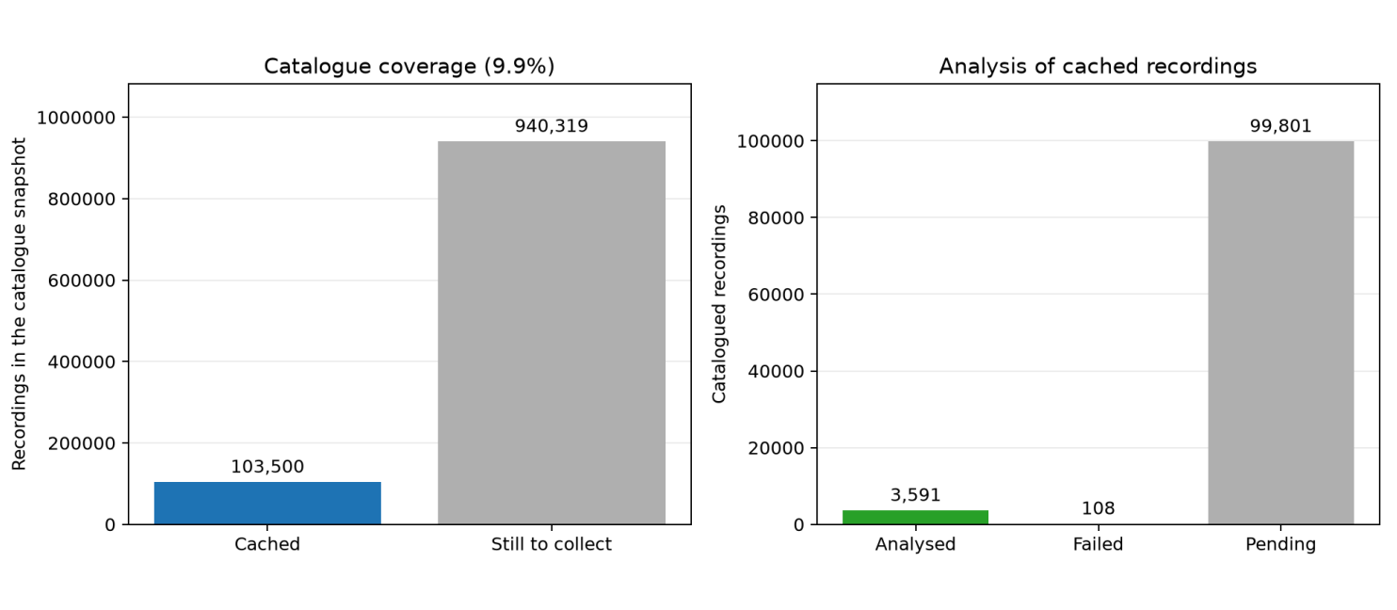
The additional controlled experiment selects positive clips from annotations, including initially missed species. Thresholds are selected on 25% of source recordings and evaluated on the other sources. Recorded backgrounds have no annotated bird in the reference vocabulary and come from separate calibration sources. Noise repetitions share grouped uncertainty estimates; equal-RMS mixtures retain isolated controls at the same gains. Neither this source split nor the two datasets establish geographic or training-set independence.

Data and coverage

3,591 recordings analysed · 108 failures · 7 completed batches · cursor 3,699

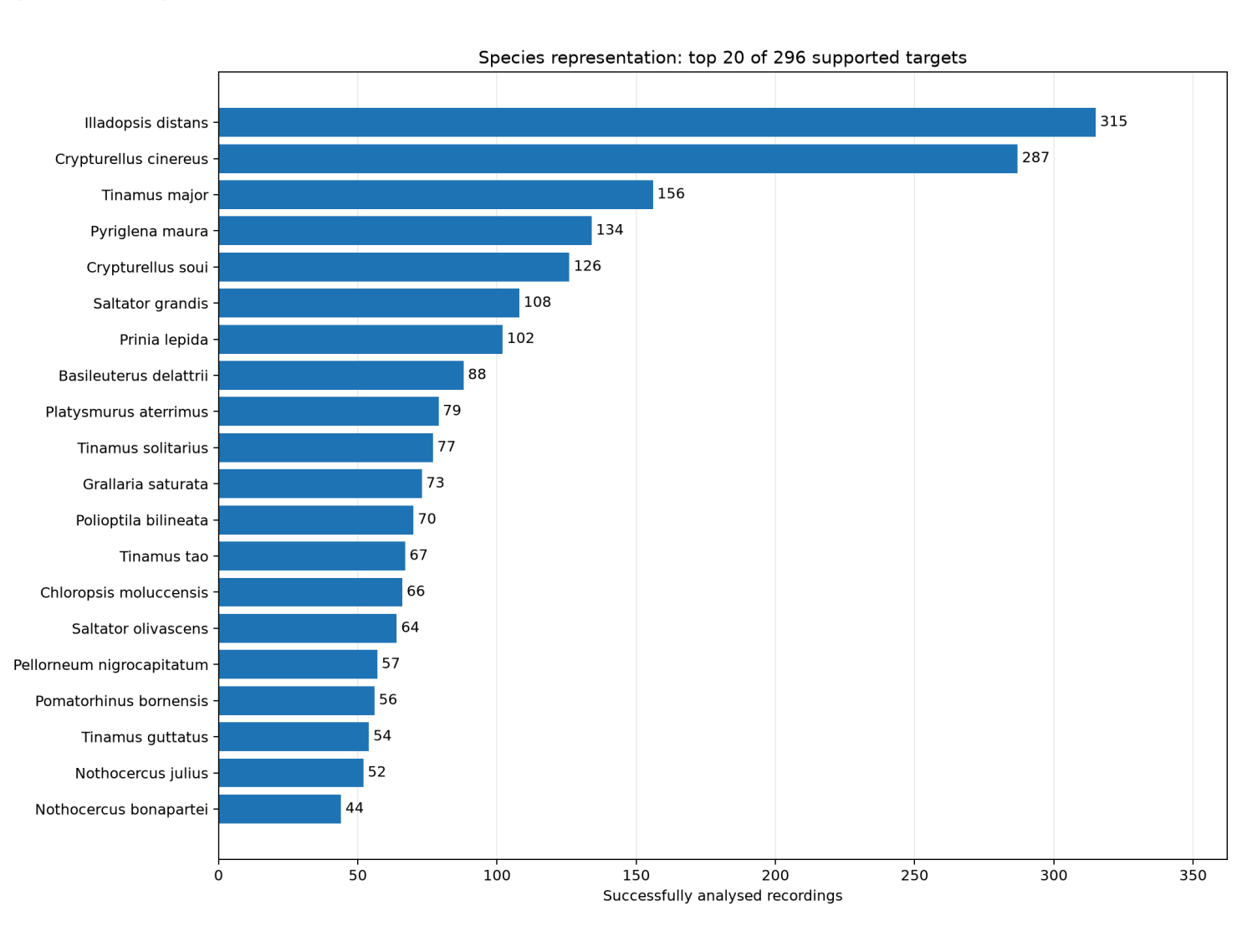
Observation. Collection is still partial; the results describe recordings already processed.

Corpus progress



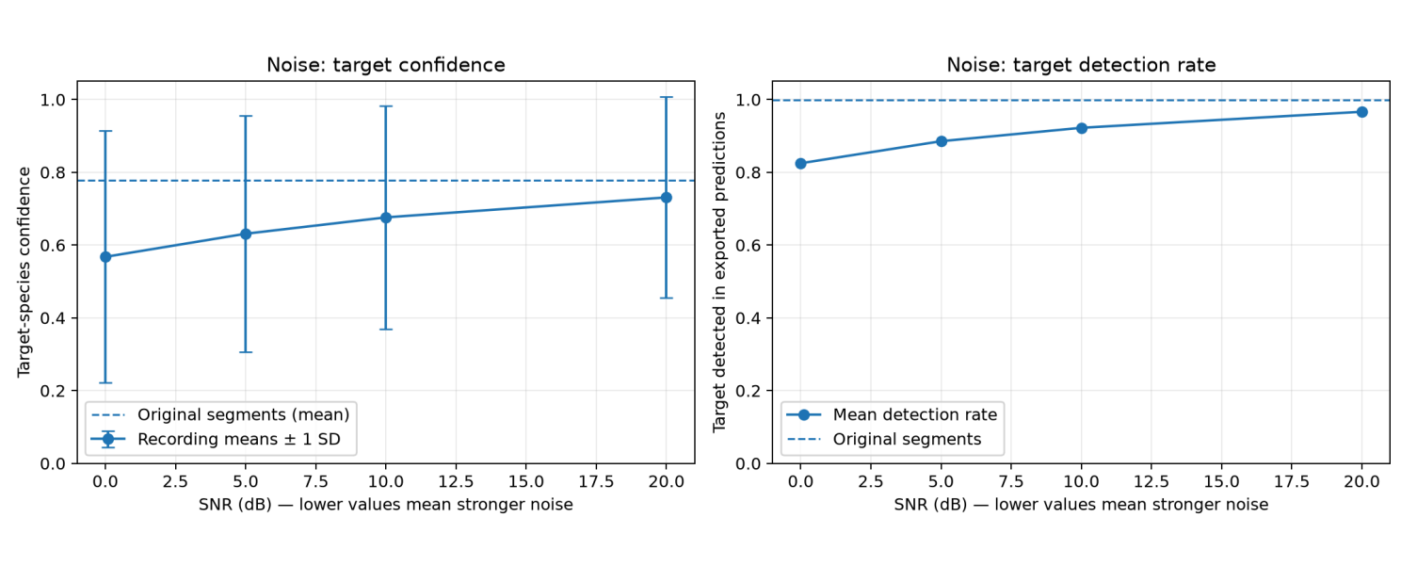
Observation. Much of the catalogue remains to be processed; the analysed corpus does not yet cover all collected recordings.

Species representation



Observation. A few species account for a large share of recordings, while many are sparsely represented.

Noise robustness



Observation. Mean confidence decreases from 0.778 to 0.568 under the strongest noise. This measurement covers initially recognised targets; 654 recordings are excluded from the noise experiment.

Noise intervals and paired effects

Noise: results per recording, 95% CI

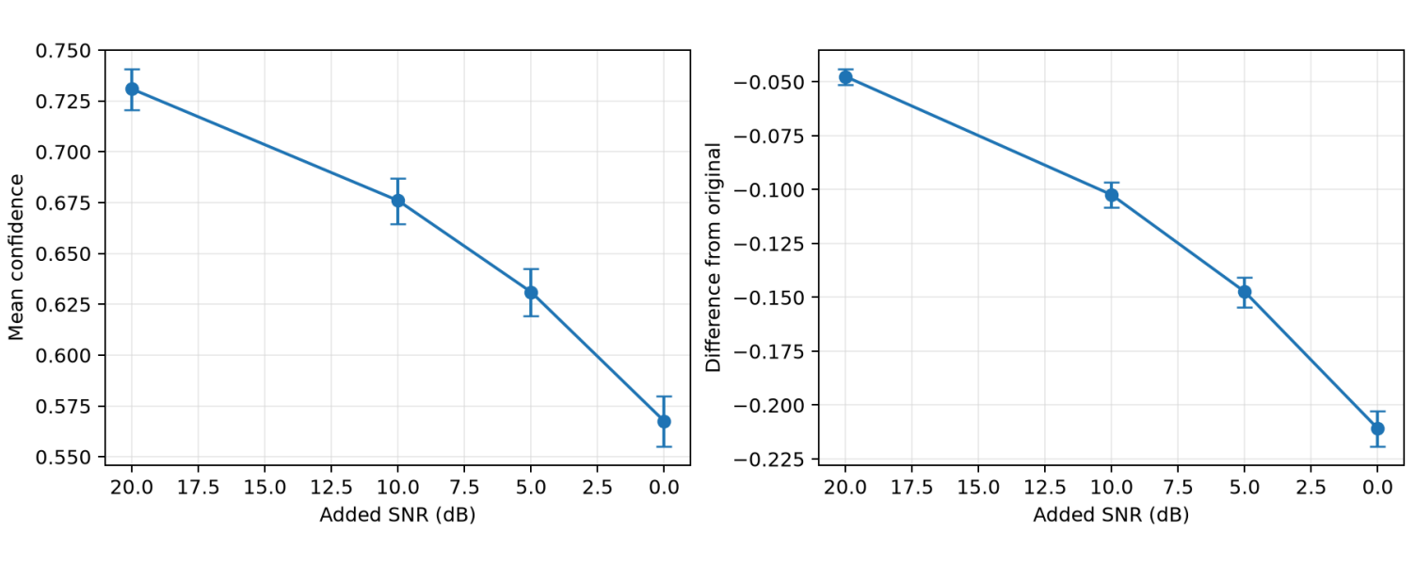
Condition	Recordings	Score [CI]	Recovery [CI]	Paired difference [CI]
original	2937	0.778 [0.770; 0.787]	0.999 [0.997; 1.000]	0.000 [0.000; 0.000]
20 dB	2937	0.731 [0.721; 0.741]	0.966 [0.960; 0.972]	-0.048 [-0.051; -0.044]
10 dB	2937	0.676 [0.664; 0.687]	0.922 [0.914; 0.931]	-0.102 [-0.108; -0.097]
5 dB	2937	0.631 [0.619; 0.643]	0.885 [0.875; 0.895]	-0.147 [-0.155; -0.141]
0 dB	2937	0.568 [0.555; 0.580]	0.825 [0.813; 0.836]	-0.211 [-0.219; -0.203]

Noise: equal species weights and clusters, 95% CI

Condition	Species	Species score [CI]	Species recovery [CI]	Species-cluster difference [CI]
original	244	0.741 [0.710; 0.770]	0.998 [0.995; 1.000]	0.000 [0.000; 0.000]
20 dB	244	0.697 [0.663; 0.730]	0.947 [0.925; 0.966]	-0.048 [-0.061; -0.039]
10 dB	244	0.637 [0.601; 0.673]	0.884 [0.850; 0.915]	-0.102 [-0.128; -0.084]
5 dB	244	0.596 [0.556; 0.635]	0.846 [0.809; 0.880]	-0.147 [-0.181; -0.123]
0 dB	244	0.537 [0.497; 0.577]	0.780 [0.737; 0.822]	-0.211 [-0.252; -0.181]

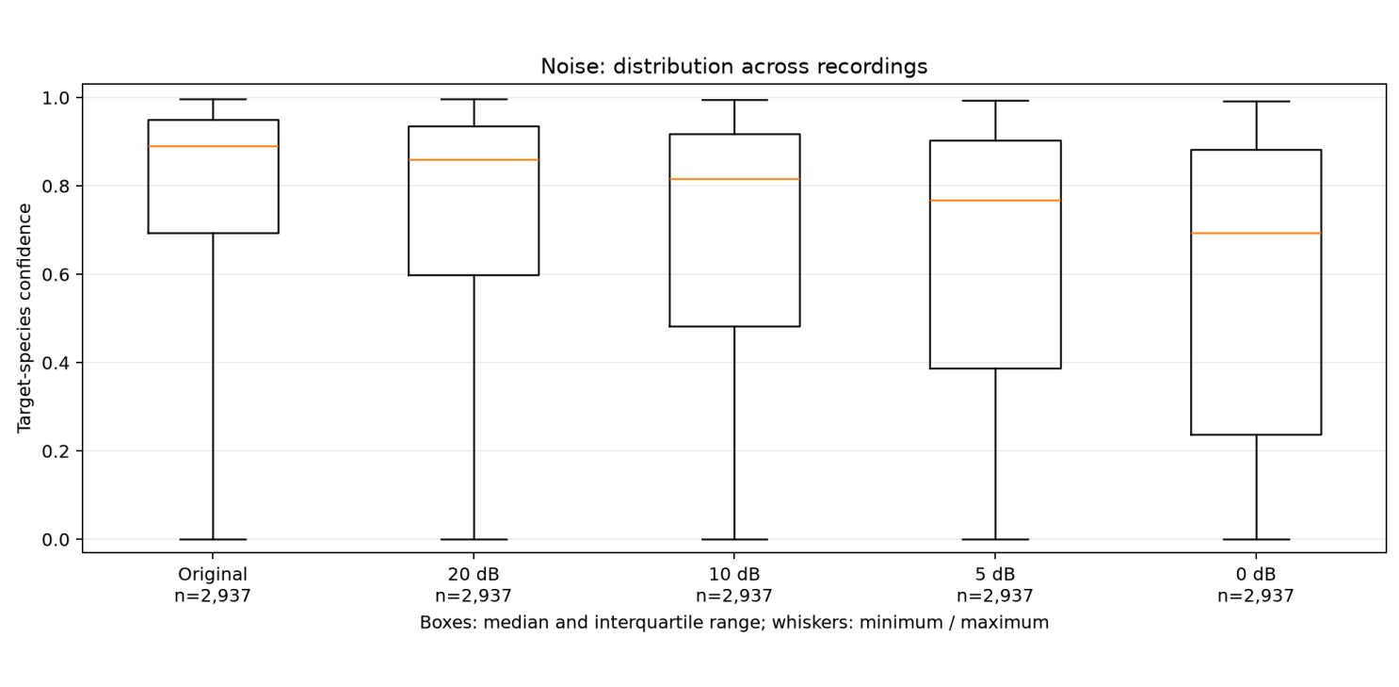
Observation. Degradation persists when species have equal weights; cluster intervals are wider.

Difference from the original score with uncertainty



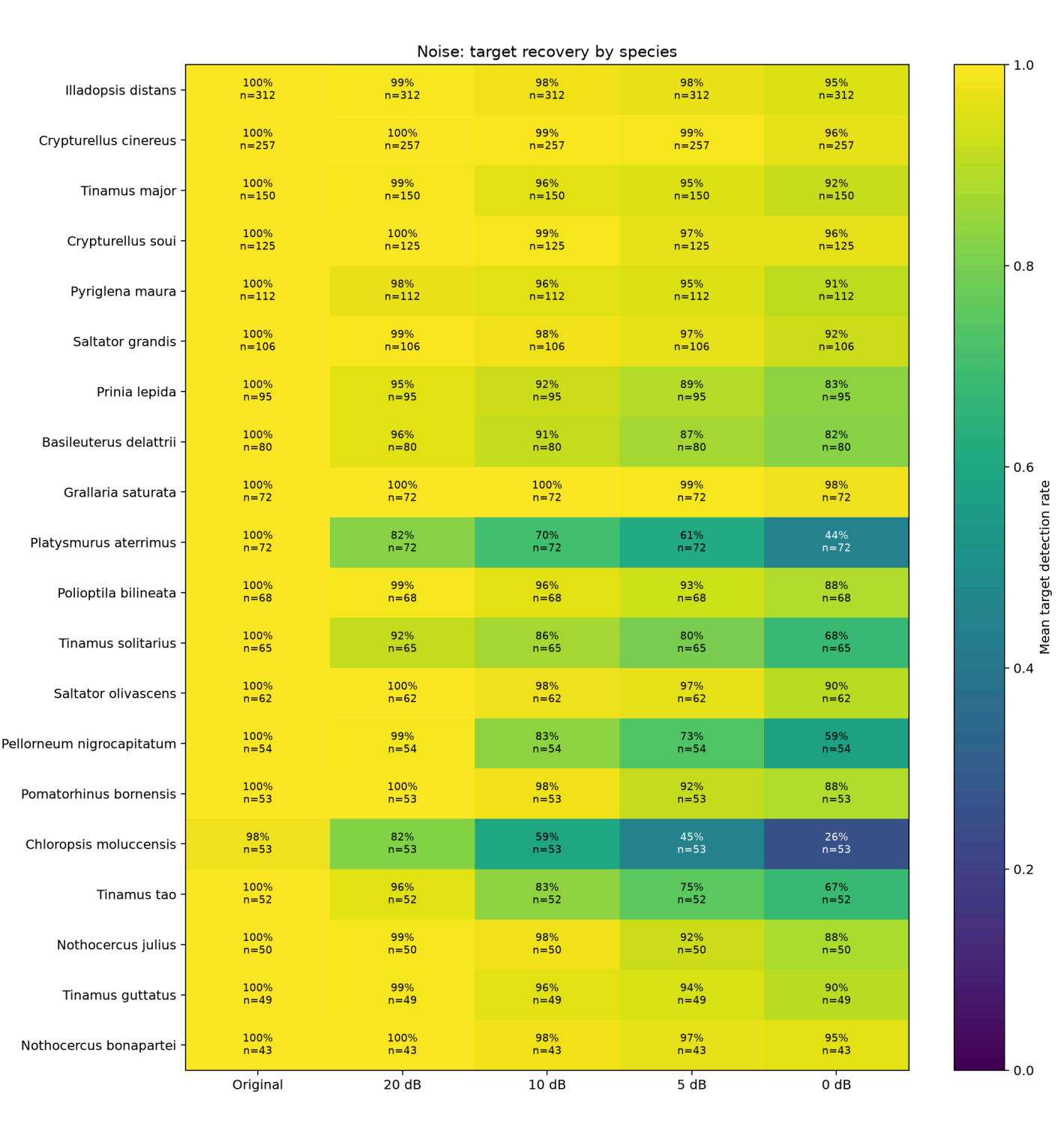
Observation. The score decrease under noise is visible with the 95% intervals.

Score distribution under noise



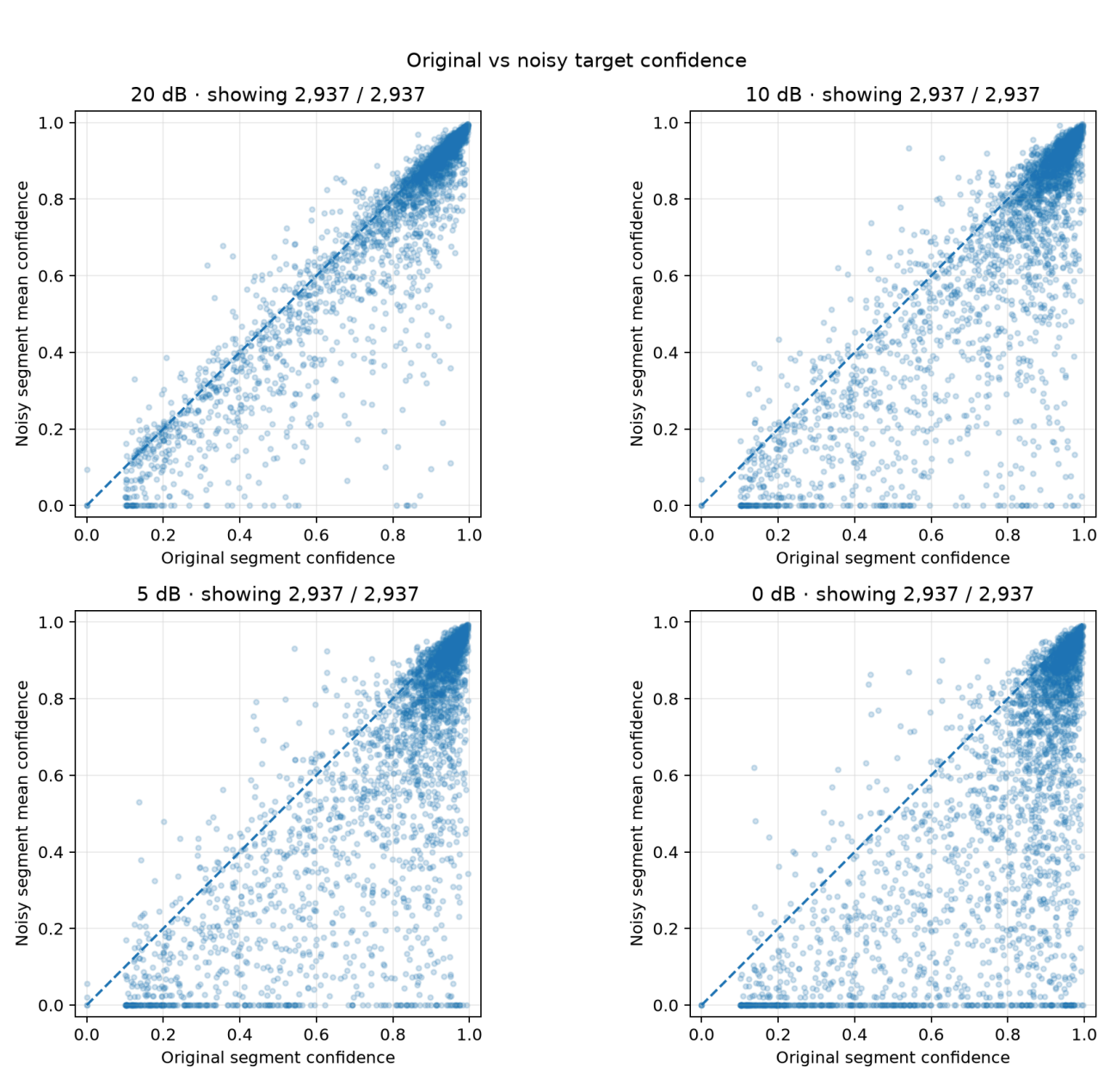
Observation. Noise shifts scores towards lower values, with substantial differences between recordings.

Species sensitivity to noise



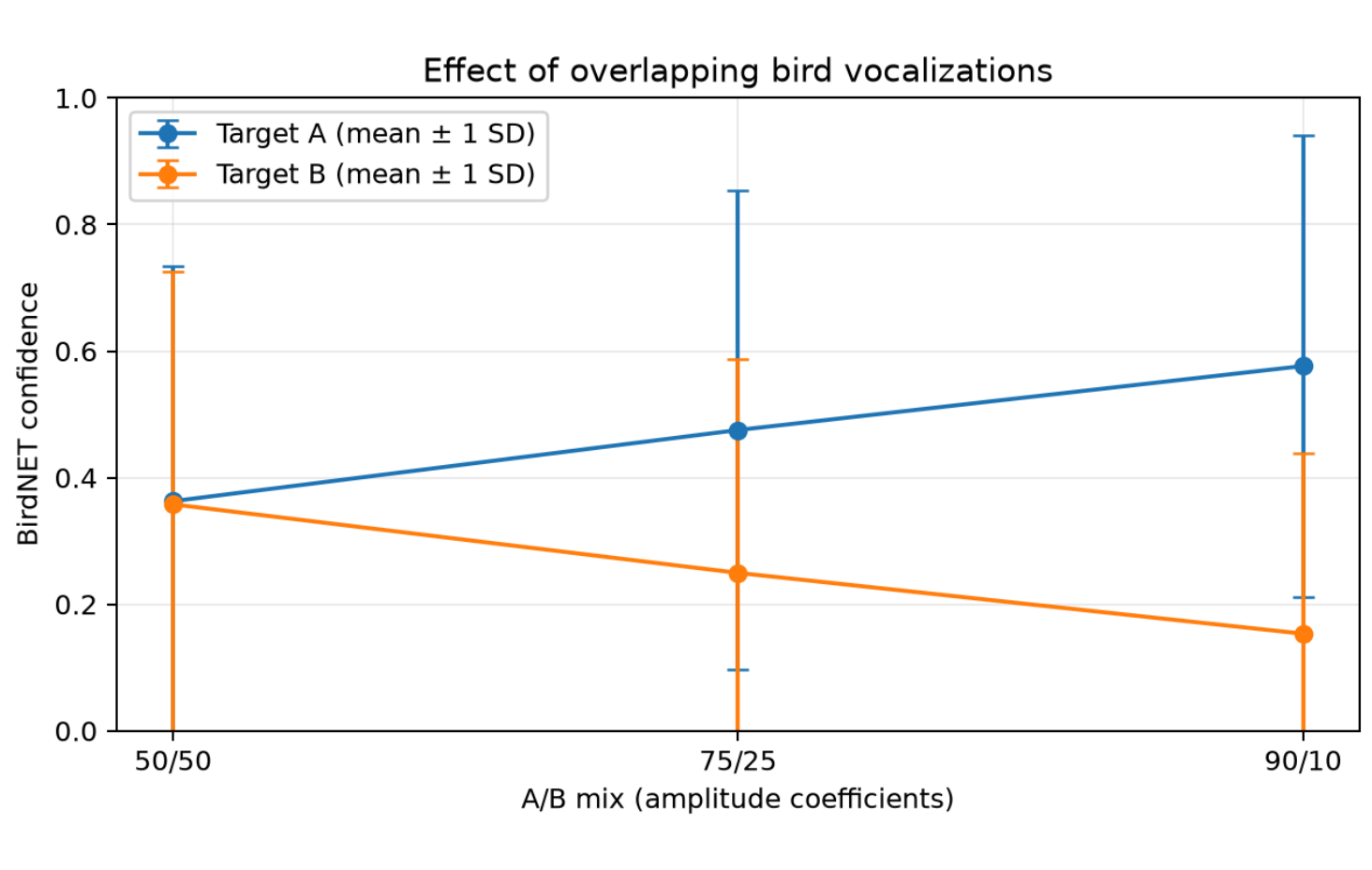
Observation. The represented species respond differently to noise; their recording counts also differ.

Original and perturbed scores



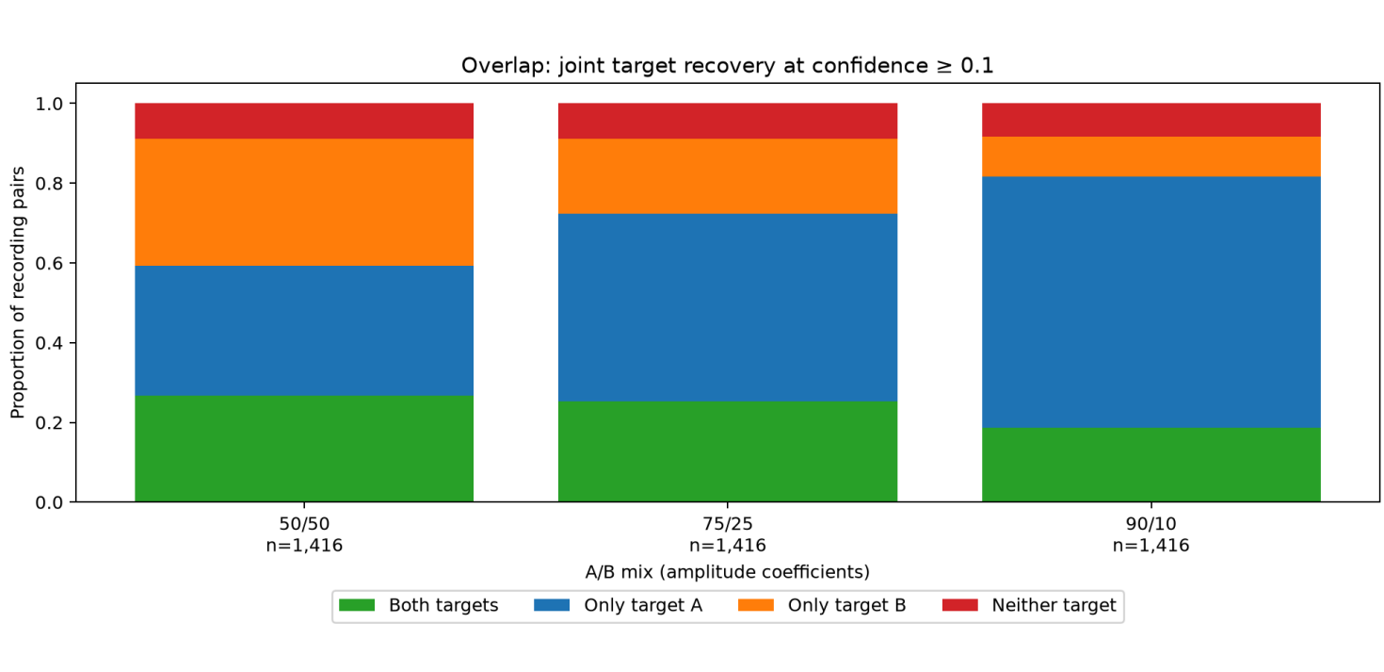
Observation. Many points fall below the diagonal as noise increases: their score becomes lower than the original.

Overlapping vocalizations



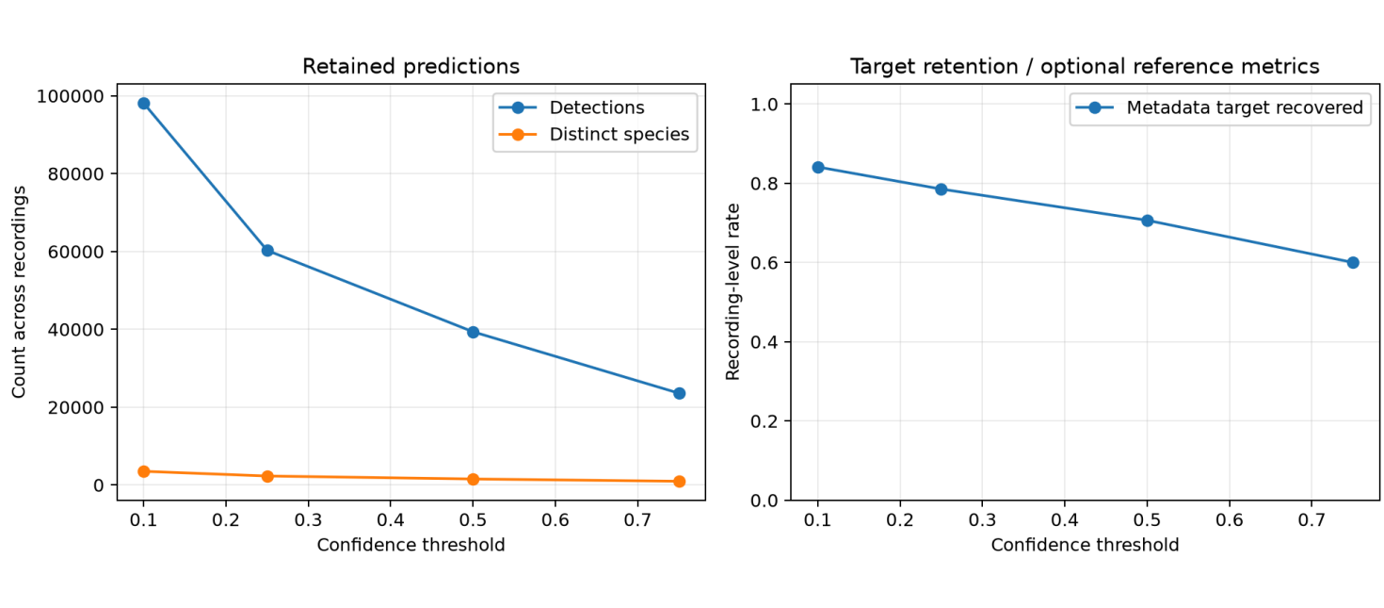
Observation. As the contribution of B decreases from 50 to 10%, its mean confidence changes from 0.358 to 0.153. The weaker target is more strongly masked.

Joint target recovery



Observation. In unbalanced mixtures, recovering only A becomes more frequent than recovering both targets.

Confidence-threshold filtering



Observation. Raising the threshold from 0.10 to 0.75 reduces retained predictions from 98,117 to 23,583. These predictions are not a validated species inventory.

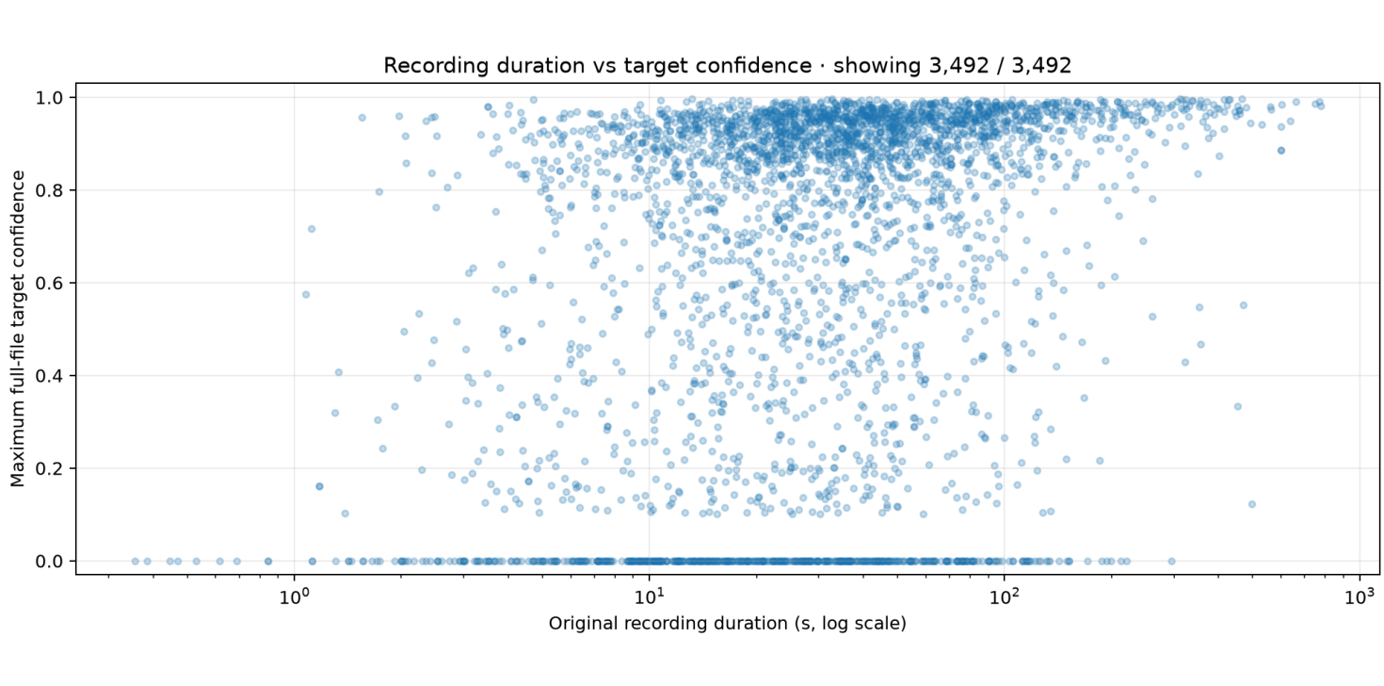
Recovery per recording and per species

Thresholds: target recovery, 95% CI

Threshold	Recordings	Per recording [CI]	Per species [CI]
0.10	3492	0.841 [0.829; 0.853]	0.751 [0.707; 0.795]
0.25	3492	0.785 [0.772; 0.799]	0.687 [0.640; 0.735]
0.50	3492	0.706 [0.692; 0.722]	0.619 [0.566; 0.670]
0.75	3492	0.600 [0.584; 0.617]	0.522 [0.471; 0.573]

Observation. Filtering reduces recovery; the species average remains below the recording average.

Recording duration and target score



Observation. Longer recordings tend to have a higher maximum score; they also provide more analysis windows.

Descriptive results by country

Recovery at threshold 0.10, groups with at least 30 recordings

Country	Recordings	Species	Recovery [CI]
Brazil	365	33	0.844 [0.805; 0.879]
Ecuador	351	26	0.940 [0.915; 0.963]
Colombia	339	27	0.844 [0.805; 0.882]
Tanzania	332	7	0.973 [0.955; 0.988]
Peru	239	22	0.916 [0.883; 0.950]
Malaysia	229	11	0.886 [0.847; 0.926]
Indonesia	150	28	0.640 [0.560; 0.713]
Mexico	118	15	0.949 [0.907; 0.983]
Bolivia	105	11	0.857 [0.790; 0.924]
Saudi Arabia	84	7	0.738 [0.643; 0.833]
Philippines	71	10	0.761 [0.662; 0.859]
Costa Rica	70	8	0.957 [0.900; 1.000]
Japan	62	3	0.839 [0.742; 0.919]
United States	57	12	0.737 [0.614; 0.842]
French Guiana	50	5	0.980 [0.940; 1.000]
Australia	49	9	0.082 [0.020; 0.163]
Venezuela	45	9	0.956 [0.889; 1.000]
Panama	43	6	1.000 [1.000; 1.000]
Argentina	41	10	0.756 [0.610; 0.878]
New Zealand	39	4	0.744 [0.590; 0.872]
United Kingdom	31	19	0.935 [0.839; 1.000]

Observation. Rates vary between countries, but species and conditions also differ.

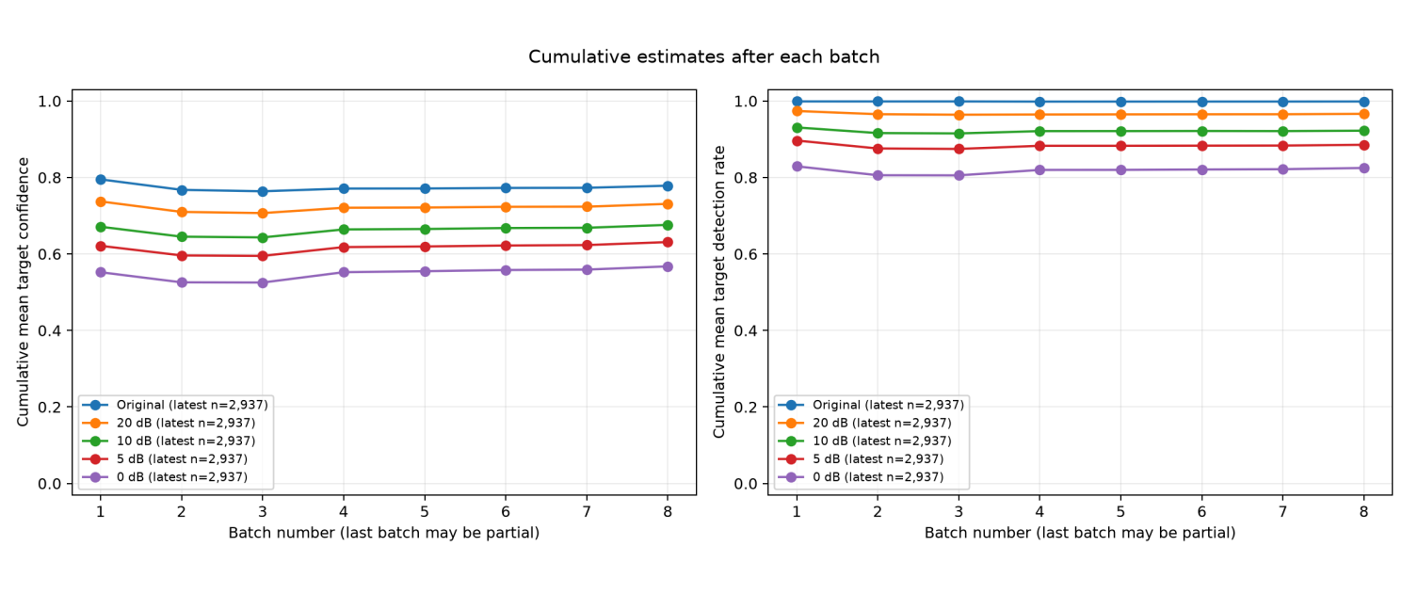
Descriptive results by quality

Recovery at threshold 0.10, groups with at least 30 recordings

Xeno-canto quality	Recordings	Species	Recovery [CI]
B	1446	195	0.843 [0.824; 0.862]
A	1060	144	0.899 [0.880; 0.918]
C	717	139	0.778 [0.748; 0.809]
D	184	61	0.777 [0.717; 0.832]
E	54	24	0.685 [0.556; 0.815]
no score	31	11	0.871 [0.742; 0.968]

Observation. Quality categories have different rates; this comparison remains descriptive.

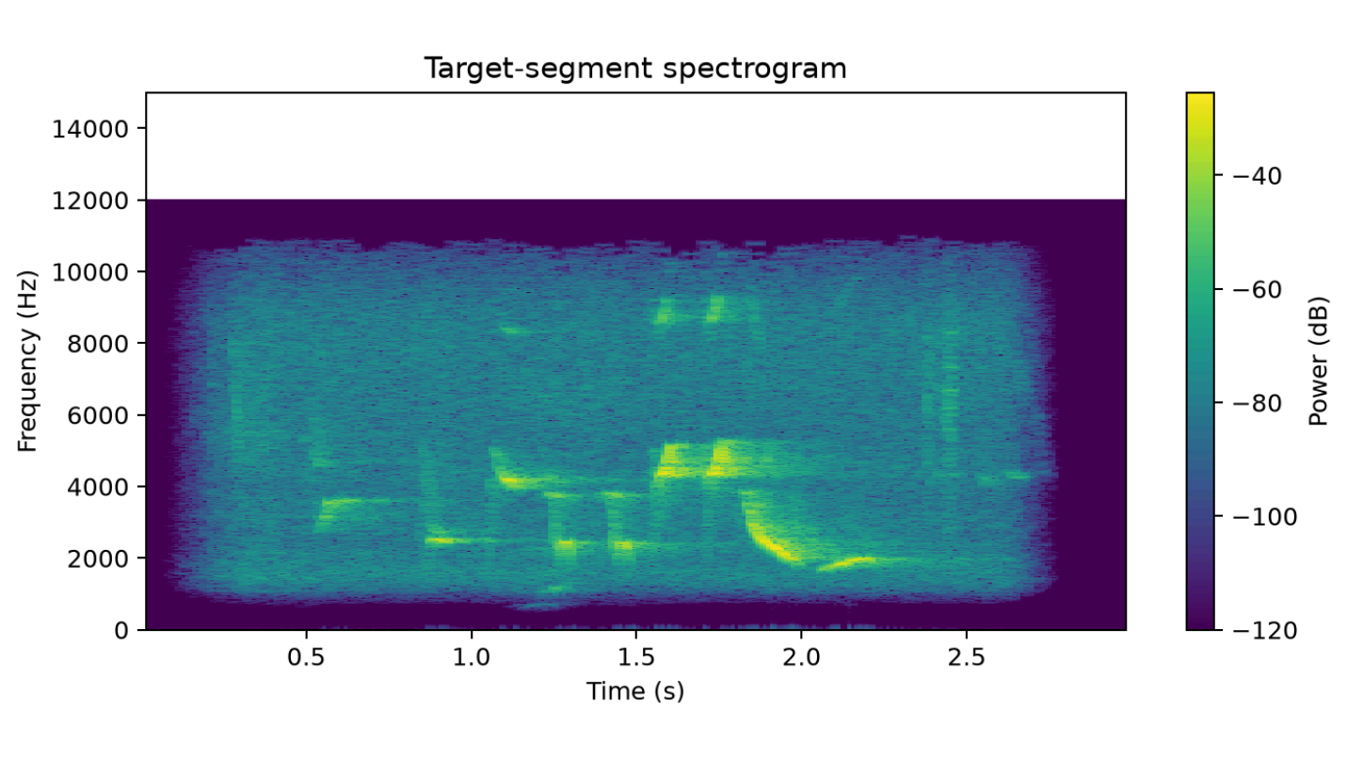
Cumulative changes by batch



Observation. Estimates change as recordings are added, and the 0 dB condition remains the most difficult.

Spectrogram

Example of an analysed segment; it does not provide a reference annotation.



Observation. The spectrogram shows the time-frequency distribution of a segment; it does not validate its species.

Multilabel soundscape classification

NBP: precision, recall and F1, all_reference_classes

Version	Threshold	Precision	Recall	Micro F1 [CI]	Macro F1
2.4	0.10	0.744	0.246	0.370 [0.336; 0.401]	0.341
2.4	0.25	0.843	0.180	0.296 [0.263; 0.331]	0.262
2.4	0.50	0.950	0.125	0.220 [0.187; 0.256]	0.191
2.4	0.75	0.987	0.080	0.148 [0.119; 0.178]	0.126
3.0	0.10	0.830	0.575	0.679 [0.651; 0.706]	0.658
3.0	0.25	0.906	0.472	0.621 [0.591; 0.649]	0.597
3.0	0.50	0.958	0.342	0.504 [0.470; 0.537]	0.486
3.0	0.75	0.985	0.215	0.352 [0.317; 0.385]	0.333

NBP: clip/species counts and false alarms

Version	Threshold	TP	FP	FN	False alarms on negative clips
2.4	0.10	227	78	696	0.0%
2.4	0.25	166	31	757	0.0%
2.4	0.50	115	6	808	0.0%
2.4	0.75	74	1	849	0.0%
3.0	0.10	531	109	392	0.0%
3.0	0.25	436	45	487	0.0%
3.0	0.50	316	14	607	0.0%
3.0	0.75	198	3	725	0.0%

NBP: evaluated vocabulary counts

Clips	Sources	Classes with positives	Negative clips
539	539	51	0

HSN: precision, recall and F1, all_reference_classes

Version	Threshold	Precision	Recall	Micro F1 [CI]	Macro F1
2.4	0.10	0.313	0.864	0.459 [0.410; 0.510]	0.447
2.4	0.25	0.590	0.660	0.623 [0.566; 0.675]	0.493
2.4	0.50	0.848	0.524	0.648 [0.585; 0.699]	0.421
2.4	0.75	0.938	0.395	0.556 [0.482; 0.620]	0.374
3.0	0.10	0.812	0.434	0.565 [0.491; 0.635]	0.524
3.0	0.25	0.860	0.359	0.507 [0.430; 0.579]	0.422
3.0	0.50	0.901	0.265	0.410 [0.336; 0.480]	0.345
3.0	0.75	0.966	0.184	0.310 [0.230; 0.385]	0.229

HSN: clip/species counts and false alarms

Version	Threshold	TP	FP	FN	False alarms on negative clips
2.4	0.10	267	587	42	66.4%
2.4	0.25	204	142	105	21.9%
2.4	0.50	162	29	147	4.8%
2.4	0.75	122	8	187	1.5%
3.0	0.10	134	31	175	2.1%
3.0	0.25	111	18	198	1.2%
3.0	0.50	82	9	227	0.9%
3.0	0.75	57	2	252	0.0%

HSN: evaluated vocabulary counts

Clips	Sources	Classes with positives	Negative clips
600	100	13	333

Observation. 3.0 achieves a higher F1 on NBP; on HSN, the result depends on the threshold.

Comparison on common NBP species

NBP: precision, recall and F1, common_classes

Version	Threshold	Precision	Recall	Micro F1 [CI]	Macro F1
2.4	0.10	0.744	0.255	0.380 [0.346; 0.412]	0.348
2.4	0.25	0.843	0.187	0.305 [0.271; 0.341]	0.267
2.4	0.50	0.950	0.129	0.227 [0.193; 0.264]	0.195
2.4	0.75	0.987	0.083	0.153 [0.122; 0.184]	0.129
3.0	0.10	0.824	0.575	0.678 [0.648; 0.705]	0.657
3.0	0.25	0.905	0.484	0.631 [0.601; 0.659]	0.604
3.0	0.50	0.958	0.355	0.518 [0.483; 0.551]	0.495
3.0	0.75	0.985	0.222	0.363 [0.327; 0.396]	0.340

NBP: clip/species counts and false alarms

Version	Threshold	TP	FP	FN	False alarms on negative clips
2.4	0.10	227	78	663	13.3%
2.4	0.25	166	31	724	6.7%
2.4	0.50	115	6	775	0.0%
2.4	0.75	74	1	816	0.0%
3.0	0.10	512	109	378	6.7%
3.0	0.25	431	45	459	6.7%
3.0	0.50	316	14	574	0.0%
3.0	0.75	198	3	692	0.0%

NBP: evaluated vocabulary counts

Clips	Sources	Classes with positives	Negative clips
539	539	50	15

Observation. The advantage of 3.0 on NBP remains visible for the 50 common classes. The 15 negative clips here are negative only within this reduced vocabulary.

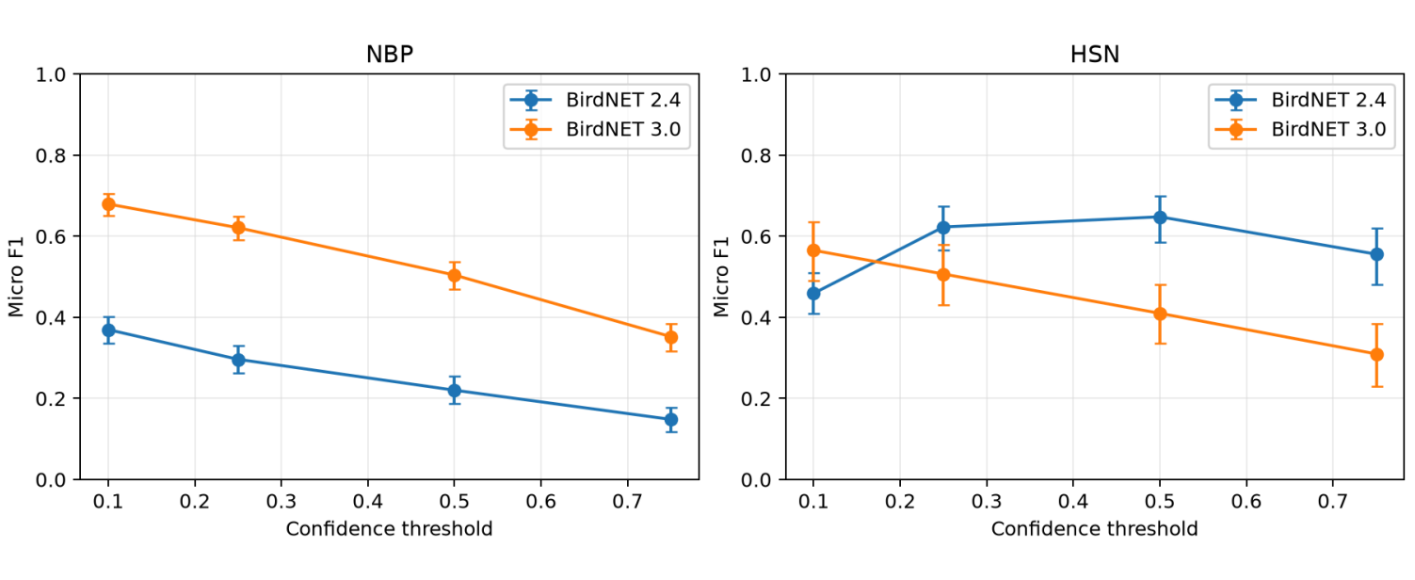
Paired model comparison

Paired model comparison, 95% CI

Dataset	Threshold	F1 difference (3.0 - 2.4)	95% CI
NBP	0.10	+0.310	[+0.277; +0.344]
NBP	0.25	+0.325	[+0.291; +0.359]
NBP	0.50	+0.284	[+0.248; +0.319]
NBP	0.75	+0.204	[+0.173; +0.236]
HSN	0.10	+0.106	[+0.026; +0.182]
HSN	0.25	-0.116	[-0.193; -0.040]
HSN	0.50	-0.238	[-0.303; -0.177]
HSN	0.75	-0.246	[-0.328; -0.170]

Observation. The difference favours 3.0 on NBP and changes sign with the threshold on HSN.

F1 by dataset and threshold



Observation. The model ranking depends on the dataset and decision threshold.

Threshold calibration on separate sources

Source validation

Dataset	Version	Threshold	Sources (calibration / test)	Precision	Recall	Micro F1 [95% CI]
NBP	2.4	0.05	135 / 404	0.669	0.311	0.425 [0.388; 0.461]
NBP	3.0	0.05	135 / 404	0.695	0.636	0.664 [0.633; 0.694]
HSN	2.4	0.50	25 / 75	0.849	0.533	0.655 [0.589; 0.716]
HSN	3.0	0.05	25 / 75	0.741	0.508	0.603 [0.533; 0.667]

Paired F1 after separate calibration

Dataset	F1 difference (3.0 - 2.4) [95% CI]
NBP	+0.239 [+0.200; +0.278]
HSN	-0.052 [-0.123; +0.016]

Observation. The threshold is chosen on calibration sources; these F1 scores use different test sources. Intervals condition on that chosen threshold.

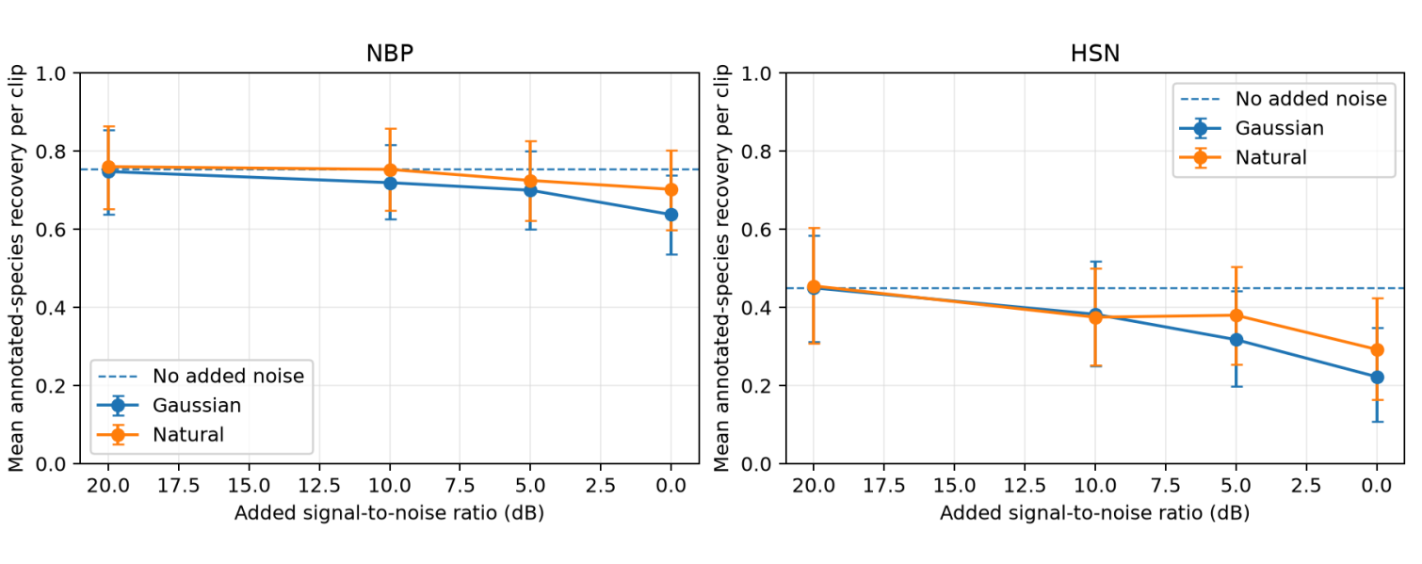
Annotation-selected targets under controlled noise

Controlled noise

Dataset	Condition	Units	Groups	Recovery [95% CI]	Paired difference (points) [CI]
NBP	clean	40	20	75.5% [64.2%; 86.4%]	+0.0 [+0.0; +0.0]
NBP	gaussian 0 dB	40	20	63.8% [53.7%; 73.9%]	-11.8 [-19.2; -5.1]
NBP	gaussian 5 dB	40	20	70.0% [60.0%; 80.1%]	-5.5 [-12.0; -0.7]
NBP	gaussian 10 dB	40	20	71.9% [62.6%; 81.6%]	-3.6 [-9.0; +0.4]
NBP	gaussian 20 dB	40	20	74.8% [63.9%; 85.4%]	-0.8 [-2.0; +0.0]
NBP	natural 0 dB	40	20	70.2% [59.9%; 80.3%]	-5.3 [-10.3; -1.3]
NBP	natural 5 dB	40	20	72.5% [62.3%; 82.6%]	-3.0 [-6.8; +0.0]
NBP	natural 10 dB	40	20	75.3% [64.8%; 85.8%]	-0.2 [-1.6; +1.3]
NBP	natural 20 dB	40	20	76.0% [65.2%; 86.5%]	+0.5 [+0.0; +1.5]
HSN	clean	40	20	45.0% [31.2%; 60.0%]	+0.0 [+0.0; +0.0]
HSN	gaussian 0 dB	40	20	22.2% [10.8%; 34.8%]	-22.7 [-35.0; -12.0]
HSN	gaussian 5 dB	40	20	31.8% [19.8%; 44.2%]	-13.2 [-21.8; -5.5]
HSN	gaussian 10 dB	40	20	38.2% [25.0%; 51.7%]	-6.8 [-13.0; -1.7]
HSN	gaussian 20 dB	40	20	45.0% [31.2%; 58.5%]	+0.0 [-4.0; +3.5]
HSN	natural 0 dB	40	20	29.2% [16.5%; 42.5%]	-15.8 [-28.5; -4.5]
HSN	natural 5 dB	40	20	38.0% [25.5%; 50.5%]	-7.0 [-14.8; -0.5]
HSN	natural 10 dB	40	20	37.5% [25.2%; 50.0%]	-7.5 [-14.5; -2.0]
HSN	natural 20 dB	40	20	45.5% [30.8%; 60.5%]	+0.5 [-3.0; +3.8]

Observation. These clips were selected from annotations, including initially missed species. Natural backgrounds come from separate sources without annotated birds.

Gaussian and natural backgrounds



Observation. Gaussian and recorded backgrounds can affect the same annotated targets differently at the same added signal-to-noise ratio.

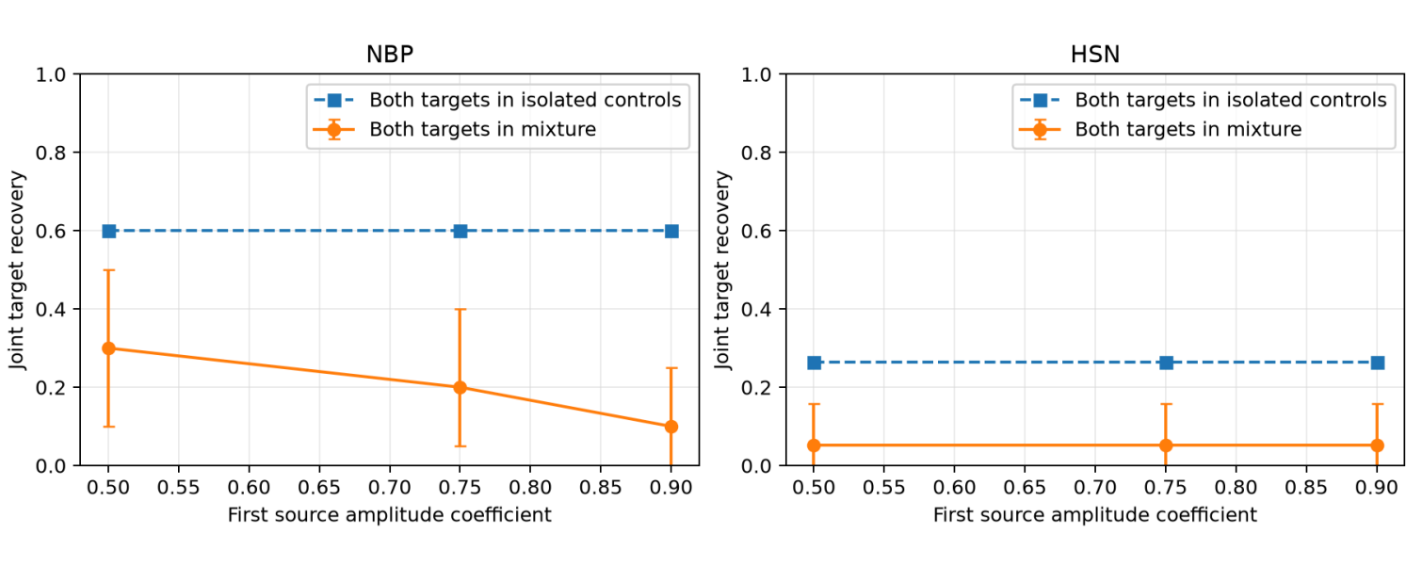
Equal-RMS mixtures with isolated controls

Controlled overlap

Dataset	Mixture	Units	Groups	Recovery [95% CI]	Paired difference (points) [CI]	Isolated controls
NBP	0.5/0.5	20	20	30.0% [10.0%; 50.0%]	-30.0 [-50.0; -10.0]	60.0%
NBP	0.75/0.25	20	20	20.0% [5.0%; 40.0%]	-40.0 [-60.0; -20.0]	60.0%
NBP	0.9/0.1	20	20	10.0% [0.0%; 25.0%]	-50.0 [-70.0; -30.0]	60.0%
HSN	0.5/0.5	19	19	5.3% [0.0%; 15.8%]	-21.1 [-42.1; -5.3]	26.3%
HSN	0.75/0.25	19	19	5.3% [0.0%; 15.8%]	-21.1 [-42.1; -5.3]	26.3%
HSN	0.9/0.1	19	19	5.3% [0.0%; 15.8%]	-21.1 [-42.1; +0.0]	26.3%

Observation. The joint recovery difference compares the mixture with its two isolated signals at identical gains; each source participates in one pair.

Joint recovery with and without superposition



Observation. Equal source RMS controls the relative input levels. Isolated controls show whether both targets were recoverable before superposition.